



Silvicultural options for rehabilitating high-graded mixedwood stands in northeastern North America

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ABSTRACT

In northeastern North America, mixed hardwood-conifer forests have commonly been harvested with selective practices such as diameter-limit cutting. By removing trees of highest commercial value, these cuts often left stands with highly variable density and reduced volume, less species diversity, and lower wood quality. The lack of care to the regenerating layer also resulted in regeneration deficiencies, especially on productive sites supporting yellow birch (*Betula alleghaniensis* Britton)-conifer stands. We assessed the 15-yr effects of four rehabilitation scenarios on stand growth, quality, vigor and regeneration in an experiment established in Quebec, Canada. The trial compared untreated experimental units (e.g. control), uniform shelterwood method (50 overstory trees/ha), strip clearcutting (20-m strips), and seed-tree method (10 overstory trees/ha), all combined with 3 site preparation treatments (no scarification, mechanical raking, and spot scarification). After 15 years, the ongoing recovery of growth, vigor and regeneration in the control and untreated part of the strip clearcut plot reflected the resilient nature of these mixedwood stands, with conifer (*Abies balsamea* [L.] Mill., *Picea rubens* Sarg., *Picea glauca* [Moench] Voss) saplings and poles contributing to the recovery of total BA and that of vigorous trees (i.e. acceptable growing stock [AGS]). The control and strip clearcut treatments, however, had high BA of unacceptable growing stock. Shelterwood plots recovered more slowly in BA and AGS, but had greater BA of high-quality trees than the control. Seed-tree method was less optimal because of its slow recovery and high understory competition. The lack of synchronization with a good seed year limited the efficiency of scarification, and increased the abundance of non-commercial species. An active rehabilitation strategy combining shelterwood system and ground disturbance (e.g. raking) during a good seed year should help to improve stand quality and regeneration. Yet, our results indicate that retaining conifers in the understory and overstory would ensure a seed source and accelerate recovery. Where seed sources and conifer advance regeneration are lacking, enrichment planting may be necessary to maintain a mixedwood composition over time.

1. Introduction

In Northeastern North America, over 10 million hectares of mixed hardwood-conifer forest mark the transition between the temperate and boreal forests (Kabrick et al., 2017). When the species mixture includes valuable hardwood species such as yellow birch (*Betula alleghaniensis* Britton), these stands have often been harvested by selective practices such as diameter-limit cutting (also called high grading) to yield the most valuable trees with a high short-term return on the investment (Prévost and Charette, 2019). Since this practice does not aim to fulfill silvicultural objectives such as improving stand growth, quality, composition and regeneration, it is not considered as silviculture (Nyland, 1992, Kenefic and Nyland, 2005, Kenefic, 2014). Financial benefits are short term because stands have a lower value and a poorer potential for

growing timber in the long term than stands managed with state-of-art silvicultural systems (Kenefic et al., 2005, Rogers et al., 2017). In addition, diameter-limit cutting alters forest ecosystems by decreasing the abundance of key attributes for wildlife and plant diversity, such as large diameter trees, large snags and down woody material (Angers et al., 2005). Impacts on residual stands vary with pre-cut stand conditions, harvest intensity and frequency (Nyland, 2016). High graded stands typically contain poor quality trees and a highly variable stocking and crown cover because of inconstant harvest intensity (Leak et al., 1987, Robitaille and Boivin, 1987, Kenefic and Nyland, 2005). They are also impoverished in terms of composition and regeneration, due to a reduced density of quality seed trees, combined to the lack of care to protect the regenerating layer. Damages by spruce budworm (*Choristoneura fumiferana*) epidemic, and expansion of non-commercial

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Table 1

ANOVA *p*-values of stand characteristics (m²/ha) by stock type, diameter size and group of species with repeated measurements at years 0, 5, 10 and 15 after treatment.

	Total merchantable basal area			Acceptable growing stock (AGS)			High-quality stock		
	ndf	ddf	<i>p</i> > <i>F</i>	ndf	ddf	<i>p</i> > <i>F</i>	ndf	ddf	<i>p</i> > <i>F</i>
Cutting	3	9.21	< 0.001	3	12.13	< 0.001	3	13.57	< 0.001
Year	3	34.21	< 0.001	3	34.58	< 0.001	3	21.71	0.354
Cut × Year	9	34.15	0.078	9	34.53	0.450	9	24.50	0.407
	Unacceptable growing stock (UGS)			AGS 10–22 cm			AGS 24 cm +		
	ndf	ddf	<i>p</i> > <i>F</i>	ndf	ddf	<i>p</i> > <i>F</i>	ndf	ddf	<i>p</i> > <i>F</i>
Cutting	3	10.36	< 0.001	3	11.45	< 0.001	3	13.09	< 0.001
Year	3	31.86	< 0.001	3	34.16	< 0.001	3	35.46	0.075
Cut × Year	9	31.87	0.645	9	34.09	< 0.001	9	35.43	0.266
	AGS birches			AGS balsam fir			AGS spruces		
	ndf	ddf	<i>p</i> > <i>F</i>	ndf	ddf	<i>p</i> > <i>F</i>	ndf	ddf	<i>p</i> > <i>F</i>
Cutting	3	9.27	< 0.001	3	10.02	< 0.001	3	12.17	< 0.001
Year	3	27.13	0.003	3	35.73	< 0.001	3	35.78	< 0.001
Cut × Year	9	28.55	0.974	9	35.70	< 0.001	9	35.77	0.005

Note: ndf = numerator degrees of freedom; ddf = denominator degrees of freedom. We used Gamma distribution in PROC GLIMMIX in all analyses and Ar(1) covariance structure when possible (arh(1) for AGS and high-quality stock).

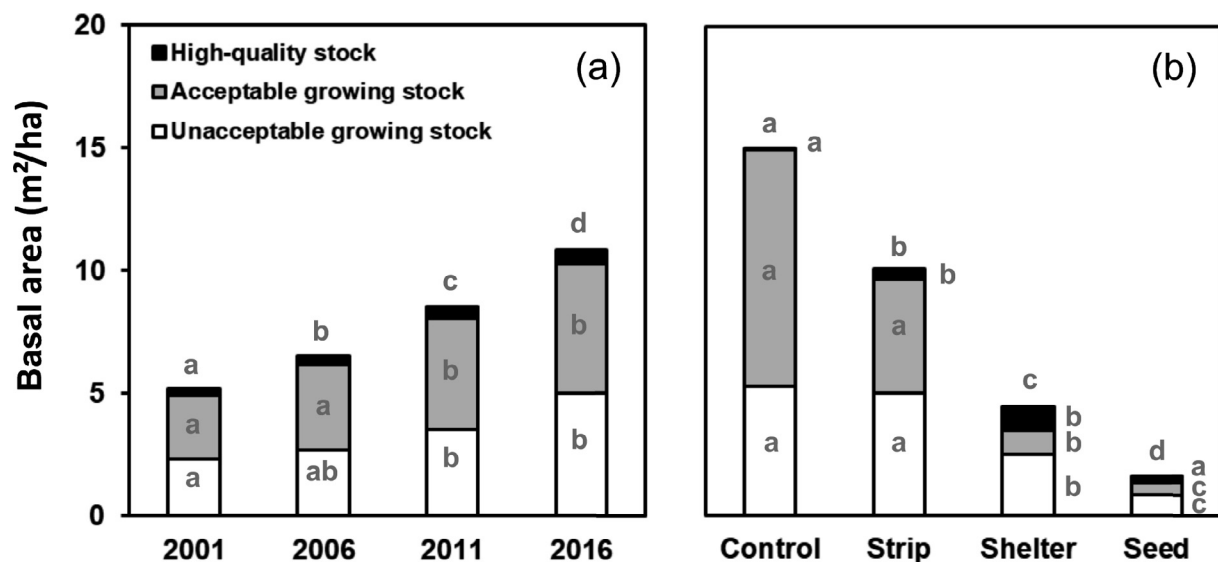


Fig. 1. Single effects of (a) year and (b) cutting treatment on basal area (m²/ha) by stock type. Full bars represent total basal area. Analysis of acceptable growing stock included high-quality stock. For a given stock type, different letters indicate a difference (a) among years (all treatments combined) and (b) a difference among cutting treatments (all years combined).

understory species and fast-growing hardwood sprouts, also contribute to further stand degradation (Kenefic et al., 2014, Power et al., 2019, Prévost and Charette, 2019). Even though diameter-limit cutting is now banned from many jurisdictions, such as in the province of Quebec, Canada, hundreds of thousands of hectares harvested in the past still require a certain degree of rehabilitation (Kenefic et al., 2014, Power et al., 2019).

Nyland (2016) defines rehabilitation as the process of reestablishing the functionality in stands compromised by a disturbance. Yet, restoring the main components that enable forest self-functioning, such as the regeneration process, is a step towards ecosystem rehabilitation (Holl and Aide, 2011). In a context of commodity production, rehabilitation implies improving timber production in terms of quantity, quality, and composition, while ensuring future crops by establishing a new cohort of desirable species. These objectives could be achieved through the

choice of rehabilitation methods that can help managing for composition and quality through the selection of trees to harvest, and ones to retain as seed sources (Nyland, 2016, Prévost and Charette, 2019).

Spatial configuration of the residual stand modulates understory light transmission and can be used to address the ecological requirements of target species while limiting the expansion of competing vegetation (Beaudet et al., 2011, Prévost and Raymond, 2012). Additionally, site preparation including vegetation control and favorable seedbed creation may prove essential to reduce interference with the regeneration of target species, especially for slow growing, shade-tolerant conifers such as red spruce [*Picea rubens* Sarg.] and balsam fir [*Abies balsamea* (L.) Mill.] (Kenefic et al., 2014, Nyland, 2016, Prévost and Charette, 2019). Indeed, when high graded stands still contain trees of target species that can serve as a seed source, natural regeneration methods (e.g. shelterwood, seed-tree, strip clearcut), combined to

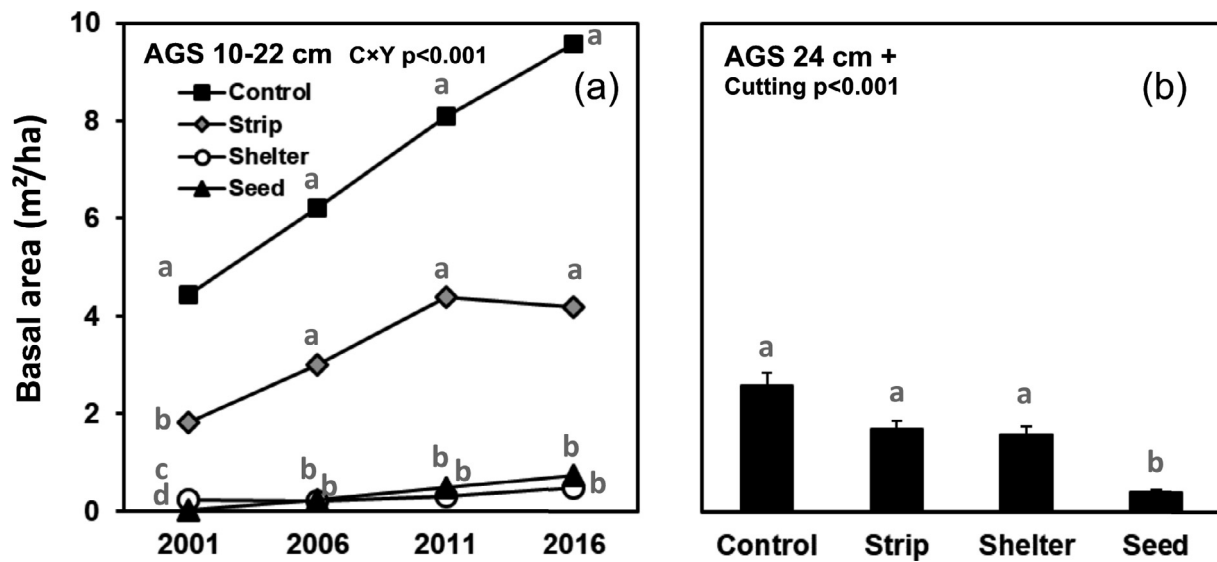


Fig. 2. Significant effects of treatments on the acceptable growing stock (AGS) by diameter classes: (a) interaction of cutting treatment $\times$ year on pole-size AGS (10–22 cm dbh classes), where different letters indicate differences among cutting treatments for a given year, and (b) single effects of cutting treatment on sawlog-size AGS (24 cm + dbh classes), where different letters show differences among treatments (all years combined). Error bars represent standard errors.

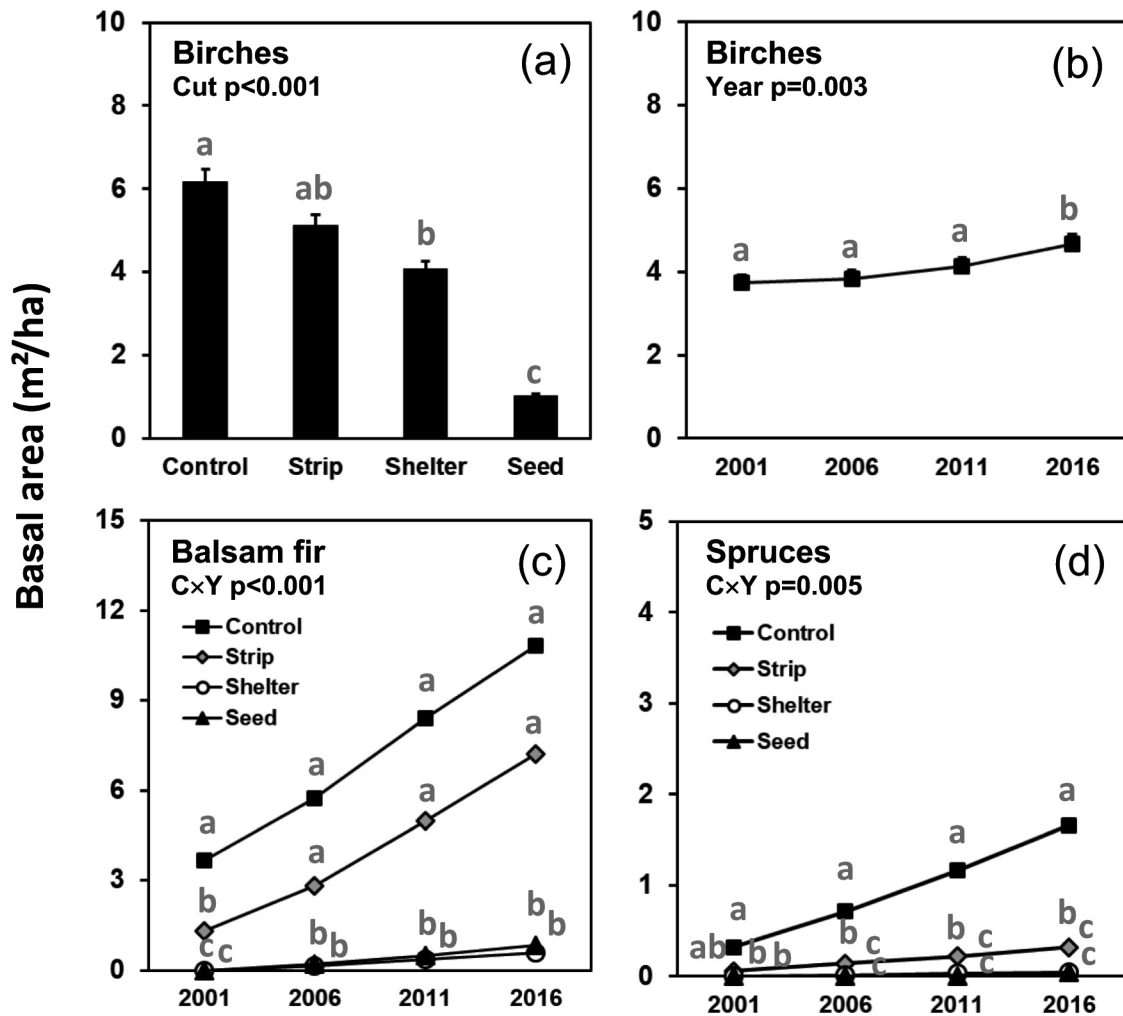


Fig. 3. Significant effects of treatments on the acceptable growing stock (AGS) by target species: (a) single effects of cutting treatment on AGS of birches, where different letters show differences among treatments (all years combined), (b) single effects of year on AGS of birches, where different letters show differences among years (all treatments combined), and finally the interaction effects of cutting treatment $\times$ year on (c) balsam fir and (d) spruces, where different letters indicate differences among cutting treatments for a given year. Error bars represent standard errors.

Table 2

ANOVA *p*-values of cutting and scarification treatments effects on seedlings (1–3 m, ≤ 1.0 cm dbh) and saplings density (1.1–9.0 cm dbh) with repeated measurements at years 1, 5, 10 and 15 after treatment.

	Seedlings			Saplings		
	ndf	ddf	<i>p</i> > <i>F</i>	ndf	ddf	<i>p</i> > <i>F</i>
Conifers						
Cutting	3	176	0.282	3	172	0.016
Scarification	2	176	< 0.001	2	172	< 0.001
Cut × Scar	6	176	0.016	6	172	0.469
Year	3	176	< 0.001	3	172	0.005
Covariate	1	176	0.002	1	172	0.001
Birches						
Cutting	3	146	0.455	3	157	0.004
Scarification	2	146	0.666	2	157	< 0.001
Cut × Scar	6	146	0.217	6	157	0.119
Year	3	146	< 0.001	3	147.6	< 0.001
Cut × Year	9	146	< 0.001	9	149.9	0.031
Scar × Year	6	146	< 0.001	6	149.4	0.030
Covariate			ns	1	157	< 0.001
Other hardwoods						
Cutting	3	158	0.771	3	149	0.333
Scarification	2	158	0.444	2	149	0.003
Cut × Scar	6	158	0.001	6	149	0.053
Year	3	158	< 0.001	3	104	< 0.001
Cut × Year	9	158	< 0.001	9	108.3	< 0.001
Scar × Year	6	158	< 0.001	6	107.3	0.015
Covariate			ns	1	149	0.009
Non-commercial						
Cutting	3	158	< 0.001	3	149	< 0.001
Scarification	2	158	0.364	2	149	0.260
Cut × Scar	6	158	0.291	6	149	0.997
Year	3	158	< 0.001	3	149	< 0.001
Cut × Year	9	158	< 0.001	9	149	< 0.001
Scar × Year	6	158	< 0.001	6	149	< 0.001
Covariate			ns	1	149	< 0.001

Notes: ndf = numerator degrees of freedom; ddf = denominator degrees of freedom; ns = non-significant. We removed the outliers for the birches and other hardwoods seedlings and saplings, as well as for the non-commercial seedlings. All analyses used a Gaussian distribution in PROC GLIMMIX and Arh (1) covariance structure.

scarification, could be used to establish a new cohort of regeneration (Raymond et al., 2009, Bédard et al., 2014, Prévost and Charette, 2019).

This paper assesses the effects of four rehabilitation scenarios on stand growth, quality, vigor and regeneration 15 years after their application in temperate mixedwood stands of Quebec, Canada. We compared untreated experimental units (e.g. control) and ones managed with 3 regeneration methods (seed-tree, strip clearcut and shelterwood), combined to 3 sites preparations (no scarification, raking and spot scarification). We predicted that units managed with these regeneration methods would significantly improve growth, quality, vigor and regeneration compared to the control. We posed the shelterwood method should offer the best rehabilitation potential because of its greater flexibility for selecting trees to remove, which should improve overall stand quality and vigor. This method is also known for leaving the best trees as seed source, which should provide favorable microclimatic conditions for a new cohort of target species (i.e. yellow birch, red spruce, white spruce [*Picea glauca* (Moench) Voss] and balsam fir). We also expected that scarification treatments should improve regeneration of small-seeded species (i.e. yellow birch and spruces).

2. Materials and methods

2.1. Study site

The study was established in 2001 in high-graded yellow birch-

conifer stands north of La Tuque, Quebec, Canada (47°49'N, 72°36'W), in the ecoregion 4c (Collines du Moyen-Saint-Maurice) of the balsam fir–yellow birch bioclimatic domain (Gosselin, 2002). Rolling hills (mean altitude 327 m, 12% slopes) with glacial till deposits characterize the landscape (Robitaille and Saucier, 1998). The region has a 160–170 day growing season, 2.5 °C as mean annual temperature and receives 900–1000 mm of annual precipitations, with 30% falling as snow (Robitaille and Saucier, 1998). Selected stands were previously harvested by diameter-limit cutting during the fall of 1988, using respectively 34 cm, 30 cm and 20 cm as diameter at breast height (dbh, measured at 1.3 m above the ground) respectively for yellow birch, spruces and balsam fir. Prior to the rehabilitation treatments in 2001 (13 years after high grading), stands exhibited low merchantable (> 9.0 cm dbh) tree density (487 stems/ha), low BA (11.5 m²/ha), and an overstory comprised of poor form and weak yellow birch (6.5 m²/ha), balsam fir (3.0 m²/ha), paper birch (*Betula papyrifera* Marsh.) (1.5 m²/ha) and scattered white and red spruces (0.2 m²/ha). These low density stands (< 40% forest cover) contained a dense understory layer with saplings (> 1 cm dbh) of non-commercial species (6 400 stems/ha) such as mountain maple (*Acer spicatum* Lam., 3 200 stems/ha), pin cherry (*Prunus pensylvanica* L.f., 1 450 stems/ha), and beaked hazelnut (*Corylus cornuta* Marsh., 1 030 stems/ha). It also had commercial species (5 800 stems/ha) like balsam fir (2 075 stems/ha), paper birch (1 500 stems/ha), red maple (*Acer rubrum* L., 1 120 stems/ha), yellow birch (565 stems/ha) and spruces (165 stems/ha). Transmitted light was 20% at 1.50 m height (unpublished data). Vegetation grew on mesic to subhydryc soils of loamy sand to sandy loam texture and a 5-cm depth humus.

2.2. Experimental design

The split-plot design consisted of 4 completely randomized blocks comparing 4 regeneration treatments, each one applied on 1-ha (100 × 100 m) experimental units: control (i.e. let grow option, 250–400 stems/ha, 11.5 m²/ha), uniform shelterwood cutting (establishment cut leaving 50 overstory trees/ha (3.5 m²/ha), seed-tree cutting (10 overstory trees/ha), and strip clearcut (20-m wide strips). Large yellow and paper birches with well-developed crowns were preferred and marked for retention during logging in the shelterwood and seed-tree treatments. Manual operations with chainsaw felling and cable skidding were carried out from September 18th to October 18th 2001. During logging, the shelterwood, seed-tree and cut strip of the strip clearcut had their understory vegetation reduced in order to decrease competition and facilitate the establishment of a new cohort of target species. Within each experimental unit, 2 regeneration options were tested on 50 × 100 m subplots: natural regeneration and planting. This paper only focuses on the natural regeneration option, where each subunit was further divided into thirds (50 × 33 m) to test site preparation treatments: no scarification, spot scarification by an excavator (Mitsubishi 440, 2 × 5 m spots, 400 spots/ha) and mechanical raking (rake-mounted Timberjack 380). Scarification was performed during the leaf-off period between October 22nd and November 15th, 2001 with the goal to facilitate the germination of birches and spruces. These species have very small seeds and root systems that need a direct access to a steady source of moisture to survive and develop (Willis et al., 2015). Spot scarification exposed mineral horizons on small and dispersed areas, while raking mixed the organic and mineral horizons on most of the area.

2.3. Data collection and analysis

In each experimental unit, we monitored all trees > 9.0 cm dbh in the mensuration plots for diameter growth, state, vigor (Majcen et al., 1990) and stem quality (MRN, 1995). Size varied depending on treatment. In the seed-tree and shelterwood treatments, the full 1-ha was monitored, while in the control we used a 30 m × 60 m subplot,

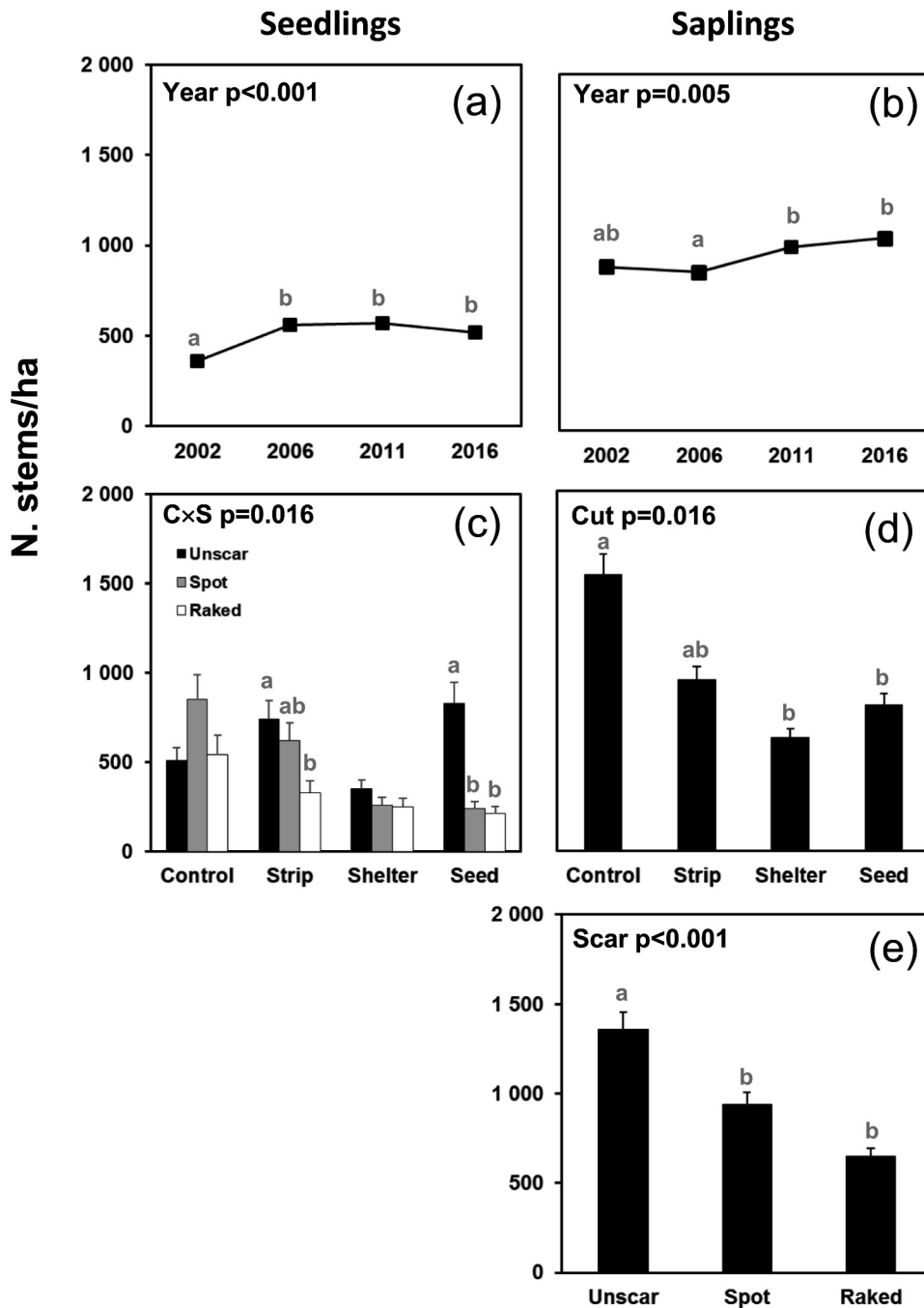


Fig. 4. Significant effects for conifer seedlings (1–3 m, ≤ 1.0 cm dbh) and saplings (1.1–9.0 cm dbh): year effects on (a) seedling and (b) sapling density, where different letters indicate a difference among years (all treatments combined), (c) cutting $\times$ scarification effect on seedling density, where different letters indicate differences among scarification treatments for each cutting treatment (all years combined), and single effects of (d) cutting and (e) scarification on sapling density, where different letters indicate respectively differences among cutting treatments (all years and scarification combined) and scarification treatments (all years and cutting treatments combined). Error bars represent standard errors.

and in the strip clearcut a 40 m $\times$ 60 m subplot that covered an equal area of 20-m of clearcut strip and 20-m of leave strip. We inventoried all seedlings (< 1 cm dbh) of woody species larger than 5 cm in height by height class (6–30 cm; 31–60 cm; 61–100 cm, 101–200 cm) in a network of 720 circular microplots (4 m², $r = 1.13$ m). We counted all woody saplings (1.1–9.0 cm dbh) by diameter class (2, 4, 6, 8 cm) in 9 m² microplots ($r = 1.69$ m), using the same centers as for the 4 m² microplots. All data were recorded prior to and immediately after the rehabilitation treatments, and at 5-yr intervals afterwards.

We assessed the 15-yr effects on stand characteristics by analyses of variance (ANOVA) of total BA, BA of vigorous trees with potential sawlog classified as I and V (Majcen et al., 1990, hereafter “acceptable growing stock” [AGS]), BA of trees unsuitable for timber production (non-vigorous and cull trees; II, III, IV and VI classes in Majcen et al., 1990, hereafter “unacceptable growing stock” [UGS]), and BA of trees

classified as A or B stem quality (MRN, 1995, hereafter “high-quality stock”). We also analyzed effects of the treatments on the AGS of poles (9.1–23.0 cm dbh, hereafter “AGS 10–22 cm”) and AGS of timber trees (> 23.0 cm dbh, hereafter “AGS 24 cm +”), using these as indicators of stand future crop trees and of sawlog volume. Then, we further evaluated the effects of the cutting treatments on the AGS of yellow birch, spruces and balsam fir.

We conducted regeneration analyses for seedlings (≤ 1.0 cm dbh, 1–3 m high) and saplings (1.1–9.0 cm dbh) of 4 different species groups: 1) conifers (balsam fir, red spruce, white spruce, tamarack [*Larix laricina* (Du Roi) Koch.], eastern white pine [*Pinus strobus* L.]); 2) birches (yellow birch, paper birch); 3) other hardwoods (red maple, sugar maple [*Acer saccharum* Marsh.], trembling aspen [*Populus tremulifolia* Michx.], large-tooth aspen [*Populus grandidentata* Michx.]) and 4) non-commercial species (pin cherry, mountain maple, striped maple [*Acer*

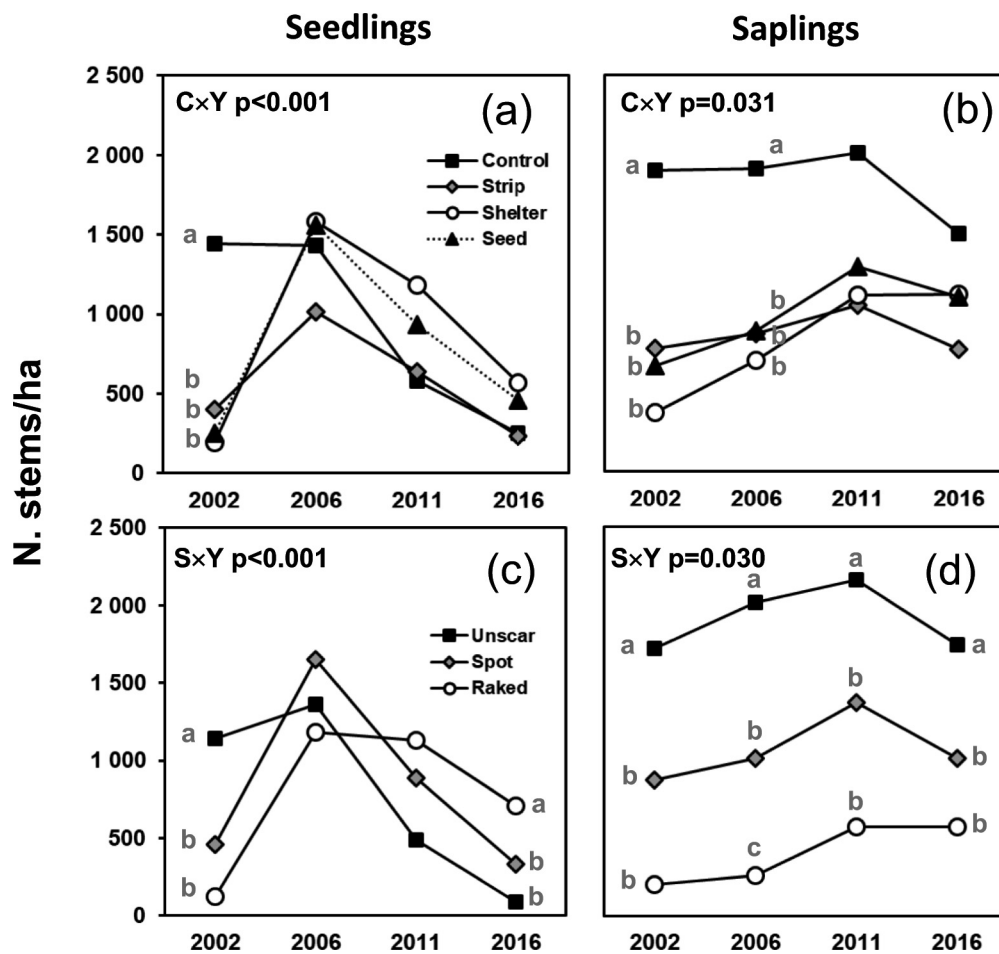


Fig. 5. Significant interaction effects for birch seedlings (1–3 m, ≤ 1.0 cm dbh) and saplings (1.1–9.0 cm dbh): cutting treatment $\times$ year on (a) seedling and (b) sapling density, where different letters indicate a difference among cutting for a given year, and scarification $\times$ year on (c) seedling and (d) sapling density, where different letters indicate a difference among scarification treatments for a given year.

pensylvanicum L.], red elderberry [*Sambucus pubens* Michx.], beaked hazelnut, American mountain ash [*Sorbus americana* Marsh.], willows [*Salix* spp.] and speckled alder [*Alnus rugosa* (Du Roi) Spreng]).

We utilized analyses of variance (ANOVAs) or covariance (ANCOVAs) to test treatment effects on stand characteristics and regeneration variables using general linear mixed models (Proc GLIMMIX; SAS 9.4, SAS Institute, Inc., Cary, NC). The analyses considered a 0.05 significance level (α), block as a random effect, and regeneration method, scarification and year as fixed effects. We used repeated measures analyses of variance rmANOVAs to examine stand characteristics and regeneration dynamics in response to treatments over time (stand variables 2001, 2006, 2011, 2016; regeneration 2002, 2006, 2011, 2016). For the regeneration, we examined changes in seedling density (individuals > 30 cm in height) using the precut (2001) values as a covariate to control for initial differences in seedling densities across plots (Milliken and Johnson, 2002). We checked the homogeneity of slopes assumption by testing the interaction between the covariate and the main effect (Milliken and Johnson, 2002). This interaction was removed from the full final model when not significant, resulting in an equal slopes model testing only the covariate. We used a Gamma distribution and a log-link function when data distributions were non-normal. We removed random effects from the model when they were near zero to avoid convergence problems (Milliken and Johnson, 1984). We estimated the treatments and year parameters effects with the LSMEAN function statement and the Tukey-Kramer adjustment for multiple comparisons and the ILINK option. We compared treatments within significant years (SLICE option) with polynomial

contrasts using the LSMEAN function statement and the SIMULATE option. Homogeneity of the variance and normality of residuals were graphically checked beforehand.

3. Results

3.1. Stand growth in BA, AGS, UGS and high-quality stock

Our results showed that over time, total BA gradually increased ($p < 0.001$), while AGS and UGS increased during the last ten years (both $p < 0.001$), and high-quality stock remained unchanged ($p = 0.354$) (Table 1, Fig. 1a). Total BA, AGS, UGS, and high-quality stock varied among cutting treatments (all $p < 0.001$, Fig. 1b). Total BA was greatest in the control ($15.0 \text{ m}^2/\text{ha}$), followed by strip clearcut ($10.0 \text{ m}^2/\text{ha}$), shelterwood ($4.4 \text{ m}^2/\text{ha}$) and seed-tree ($1.5 \text{ m}^2/\text{ha}$). We observed similar trends for the AGS and UGS, except that the strip clearcut did not differ from the control. As for high-quality stock, it was greater in the shelterwood ($1.0 \text{ m}^2/\text{ha}$) and strip clearcut ($0.4 \text{ m}^2/\text{ha}$) than in the control ($0.0 \text{ m}^2/\text{ha}$) and seed-tree ($0.3 \text{ m}^2/\text{ha}$) treatments.

We also observed differences among treatments when breaking down the AGS by diameter classes. AGS of poles (10–22 cm dbh classes) varied in interaction with cutting treatments and time ($p < 0.001$, Table 1), showing faster recovery in the control and strip clearcut than in the shelterwood and seed-tree treatments (Fig. 2a). AGS of sawlogs (24 cm + dbh classes) was higher in the control ($2.6 \text{ m}^2/\text{ha}$), strip clearcut ($1.7 \text{ m}^2/\text{ha}$) and shelterwood ($1.6 \text{ m}^2/\text{ha}$) than in the seed-tree ($0.4 \text{ m}^2/\text{ha}$) ($p < 0.001$, Fig. 2b).

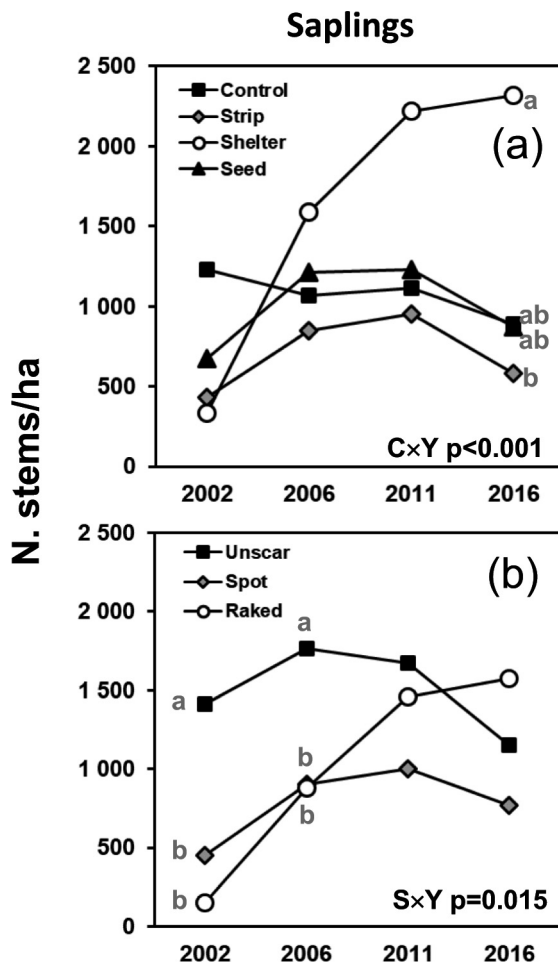


Fig. 6. Significant interaction effects for other hardwood saplings (1.1–9.0 cm dbh): (a) cutting treatment × year on sapling density, where different letters indicate a difference among cutting treatments in 2016, and (b) scarification × year on sapling density, where different letters indicate a difference among scarification treatments for a given year.

When analyzed by target species, single effects of treatments ($p < 0.001$, Table 1) showed that AGS of birches was highest in the control ($6.2 \text{ m}^2/\text{ha}$), followed by the strip clearcut ($5.1 \text{ m}^2/\text{ha}$), shelterwood ($4.0 \text{ m}^2/\text{ha}$) and seed-tree ($1.0 \text{ m}^2/\text{ha}$) (Fig. 3a). AGS of birches increased significantly in the last five years ($p = 0.003$, Fig. 3b). AGS of balsam fir increased strongly over 15 years, especially in the control and strip clearcut compared to shelterwood and seed-tree (Fig. 3c). Over the 15-yr period, AGS of spruces increased more rapidly in the control than in the other treatments (Fig. 3d).

3.2. Tree regeneration at seedling and sapling stages

Both conifer seedlings and saplings slightly increased in density with time ($p \leq 0.005$, Table 2, Fig. 4a,b). Interaction of cutting treatments with scarification ($p = 0.016$) indicates that conifer seedling density was reduced in the seed-tree treatment when the understory was treated by spot scarification or raking, and in strip clearcut when raked (Fig. 4c). Single effects showed that conifer saplings were more abundant in the control (1530 stems/ha) than in the shelterwood (630 stems/ha) and the seed-tree (810 stems/ha) treatments ($p = 0.016$, Fig. 4d), and in the unscarified plots (1360 stems/ha) than in spot-scarified (940 stems/ha) or raked (650 stems/ha) ones ($p < 0.001$, Fig. 4e).

Densities of birch seedlings and saplings varied with a combined effect of year with either the cutting or scarification treatments

($p \leq 0.031$; Table 2). Birch seedling and sapling densities were initially higher in the control than in the other treatments, but these differences diminished over the years (Fig. 5a,b). The interaction of scarification × year showed that birch seedlings were initially higher in the unscarified plots than in the other treatments, but after 15 years they were higher in the raked plots than with the other scarification treatments (Fig. 5c). Birch sapling density was generally higher in the unscarified than in spot scarified and raked plots (Fig. 5d).

Analysis detected significant cutting × scarification and cutting × year interacting effects on seedling density of other hardwood species, but further analysis did not reveal trends (Table 2). The interaction of scarification × year ($p < 0.001$) indicated an initial seedling density of other hardwoods lower in raked than in unscarified plots, but after 15 years a greater density in raked and spot scarified than in unscarified plots (not shown). Sapling density varied in interaction with cutting × year and scarification × year ($p < 0.001$ and $p = 0.015$, respectively). After 15 years, sapling density of other hardwoods was higher in shelterwood (2320 stems/ha) than in strip clearcut ($870\text{--}890 \text{ stems/ha}$) (Fig. 6a). Sapling density of other hardwoods was initially greater in unscarified plots than in scarified ones, but these differences diminished through time (Fig. 6b).

The density of non-commercial species in the seedling and sapling stages varied with year in interaction with cutting and the scarification treatments (all $p < 0.001$, Table 2, Fig. 7). Seedlings of non-commercial species were more abundant in the seed-tree than in the other treatments during all years after the first year (2002), and specifically more abundant in the shelterwood and in the strip clearcut than in the control in 2006 (Fig. 7a). Non-commercial saplings were more abundant in all treated plots compared to the control after the first year, except that the shelterwood did not differ from the control in 2006 (Fig. 7b). After 15 years, seedling density of non-commercial species was lower in unscarified subplots than in raked subplots (Fig. 7c). As for non-commercial saplings, they were initially present at higher density in unscarified plots (3330 stems/ha) than in scarified plots ($520\text{--}1060 \text{ stems/ha}$). Trends reversed by 15 years, with greater densities in scarified plots ($4320\text{--}4970 \text{ stems/ha}$) than in unscarified plots (2420 stems/ha).

4. Discussion

4.1. Rehabilitation success after 15 years

Rehabilitation success for timber production implies that the degraded components (e.g., growth, vigor, quality and regeneration) are improved over time (Nyland, 2016). Yet, our analyses revealed quite variable outcomes depending on the combination of treatments used. One striking result is the recovery of basal area and vigor (AGS) in absence of treatment, which seems largely related to development of the conifers (e.g. balsam fir) present in the lower stories at the beginning of the experiment. Indeed, diameter-limit cutting typically leaves trees smaller than the target diameters, as well as larger trees of poor quality and unsuitable for timber production (Leak et al., 1987, Robitaille and Boivin, 1987, Kenefic and Nyland, 2005). Stand density was low ($< 40\%$) in the untreated areas of the control and strip clearcut leave strip, which enabled the saplings and small stems to benefit of space and resources and contributed to the recovery of basal area and vigor (Power et al., 2019). In the shelterwood, the recovery was slower because the establishment cut reduced even more tree density (50 overstory trees/ha) and the understory was cleared of vegetation. This implies that a large part of poles, advance regeneration, seedlings and saplings were removed to enhance the environment for the new cohort. Thus, the shelterwood plots did not have an immediate growth response of conifers (Raymond et al., 2016, Power et al., 2019) as was observed in control and strip clearcut treatments. Nevertheless, the shelterwood contained fewer poor-vigor and defective trees (UGS) and more high-

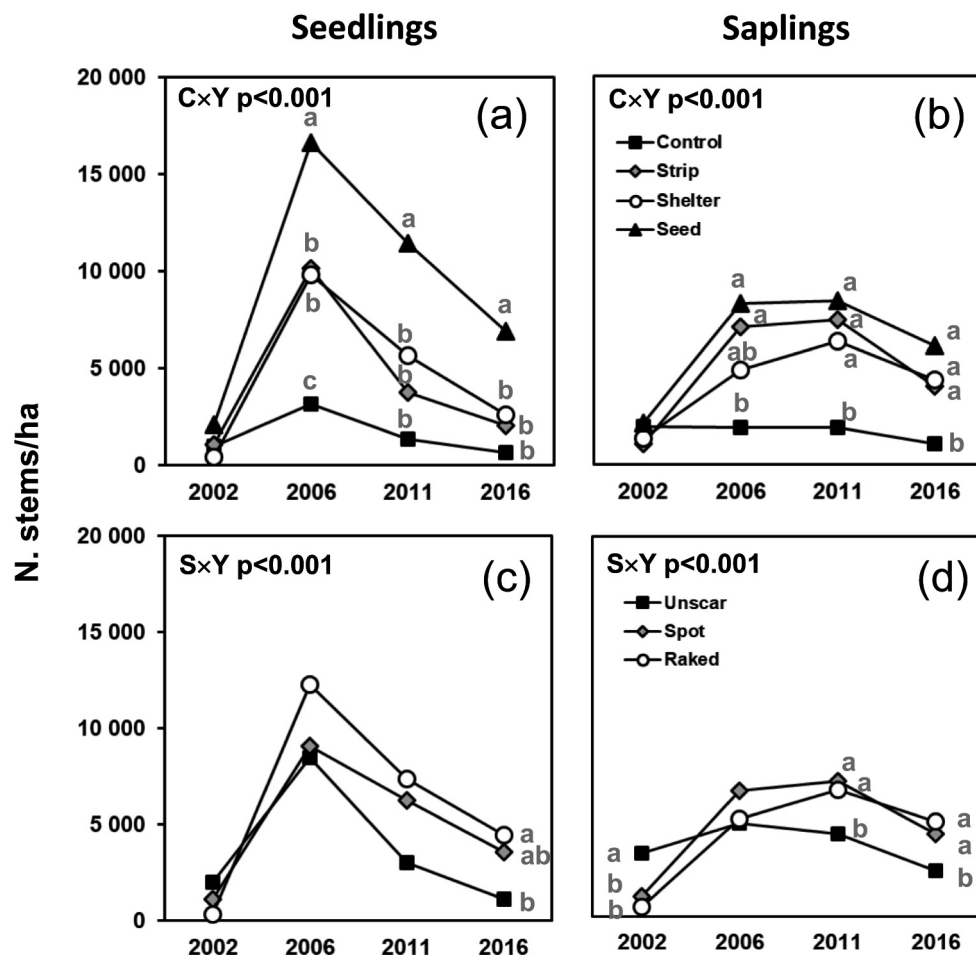


Fig. 7. Significant interaction effects for non-commercial seedlings (1–3 m, ≤ 1.0 cm dbh) and saplings (1.1–9.0 cm dbh): cutting treatment $\times$ year on (a) seedling and (b) sapling density, where different letters indicate a difference among cutting for a given year, and scarification $\times$ year on (c) seedling and (d) sapling density, where different letters indicate a difference among scarification treatments for a given year.

quality trees (grades A and B) than the control, supporting arguments that such silvicultural treatments would improve stand quality (Kenefic et al., 2014, Rogers et al., 2017). Although not the primary objective of shelterwood systems, residual trees can increase significantly in diameter growth after an establishment cut (Zarnovican et al., 2001, Sendak et al., 2003, Prévost and Dumais, 2013, Montoro Girona et al., 2016). Yet because the canopy cover was only retained on half of the area, changes within the combined clearcut strip and adjacent uncut strip proved intermediate between the control and shelterwood treatments in recovery of basal area and vigor, and also for abundance of poor-vigor and defective trees (UGS). In fact, the uncut areas of strip clearcut had a high basal area of trees unsuitable for timber production. The seed-tree cutting that retained 10 overstory trees/ha appeared the least promising with respect to short-term responses, given the longer time needed for recovery of stand basal area and abundance of high-vigor trees, and the resulting high level of understory competition.

Contrary to our predictions, the untreated units (i.e. no cutting and no scarification) provided the best regeneration outcomes for the target species (birches and conifers). These results bring to question the relevance of applying costly site preparation treatments to achieve regeneration goals, especially during poor seed years. Our study was implemented with operational scheduling constraints that led to site preparation during a poor seed year for the desired species. Thus, the newly available growing space and resources instead benefitted the non-commercial species and fast-growing hardwood sprouts. This highlights the importance of synchronizing regeneration cuts and scarification with good seed crops to maximise regeneration success

(Prévost et al., 2010, Nyland, 2016, Prévost and Charette, 2019), even given the difficulty in predicting when a good seed year will occur and common scheduling constraints.

Shelterwood systems are less sensitive to synchronization with a good seed year. They are generally recommended for regenerating birches in mixture with conifers (Holgén and Hånell, 2000, Karlsson and Nilsson, 2005, Prévost and Dumais, 2013, Raymond and Bédard, 2017), but variable results have been observed from adding the raking site preparation (Lorenzetti et al., 2008, Willis et al., 2015). Seed-tree and strip clearcut methods can also successfully establish yellow birch when combined with scarification during a good seed year, although high interspecific competition can be a problem in very low density stands (Prévost and Charette, 2019). Also, harvest and scarification operations in clearcuts can cause significant damage to advance regeneration, especially conifers (Prévost et al., 2010). The seed-tree method is also less suitable to birches that can dieback in low density stands because of their vulnerability to moisture stress (Roy et al., 2001). Further, regeneration of the last leave strips with strip clearcut method can be challenging, too, especially after the final removal (Nyland, 2016). Yet all things considered, shelterwood method would likely result in sufficient density of seedlings and saplings of target species within a new cohort, especially when combined with raking to facilitate the recruitment of new seedlings.

Our results highlight some remaining uncertainties regarding the composition of the regenerating layer, especially with respect to the proportion of conifers. Although their losses were partly compensated by establishment of new conifer seedlings, this new cohort occupies a

lower position in the understory layer, and the faster growing hardwoods may restrict conifer growth into a higher strata. Furthermore, the proportion of spruces remains low, and that aspect of the rehabilitation cannot be considered successful. Leaving spruces and other conifers as residual trees in lower and higher strata could maintain a conifer component with shelterwood and seed-tree methods, and also contribute to a faster recovery of stand basal area and vigor. Considering the scarcity of spruces to serve as seed bearers, coupled with the observed regeneration difficulties, enrichment planting may become necessary to increase the spruce component in degraded stands (e.g. Dumais and Prévost, 2019; Prévost and Charette, 2019). Results from the ongoing plantation trial in the experiment described here will eventually provide valuable information about that option.

4.2. From timber rehabilitation to ecosystem rehabilitation

Many forest ecosystems degraded by selective logging can recover passively, provided with sufficient time (Holl and Aide, 2011; Meli et al., 2017; Power et al., 2019). In fact, unmanaged natural yellow birch-conifer stands eventually transition into ones with complex uneven-aged structures (balanced or irregular, *sensu* Smith et al., 1997) after a long successional process (Hébert, 2003). However, opting for this type of passive rehabilitation (i.e. no intervention) may have an important interim opportunity cost if some goals remain unattained (Zahawi et al., 2014), at least in the short and medium terms. The recovery of volume and vigor observed in absence of any treatment indicates the resilient nature of yellow birch-conifer mixedwood stands, suggesting that intervening actively could accelerate the recovery process and improve stand quality (Kenefic et al., 2014; Rogers et al., 2017). In stands with vigorous seed bearers, regenerating with local seed sources will mimic natural succession, while plantation approaches are regarded as alternatives for stands lacking adequate supplies of seed (Parrotta, 1992; Sayer et al., 2004; Holl and Aide, 2011; Corbin and Holl, 2012). In the latter case, enrichment planting could increase the proportion of conifers to insure species diversity, especially for difficult to regenerate red and white spruces (Dumais and Prévost, 2019; Prévost and Charette, 2019). Once established, they will survive and grow under partial forest cover and in gap openings of temperate mixedwood stands (Hébert et al., 2013; Dumais et al., 2019; Dumais et al. in prep.). At the sapling stage, crop tree release could improve the performance of planted spruces (Puhlick et al., 2019; Dumais et al. in prep.). Even if not released, spruces are very shade tolerant and can act as trainer trees to improve the quality of hardwoods stems (Puhlick et al., 2019).

One further step towards ecosystem rehabilitation could include structural rehabilitation. That would begin with the choice of a regeneration method. For example, while strip clearcutting is simpler to apply, the shelterwood method offers more flexibility for future management, because the resulting two-aged stands will be more structurally complex than an even-aged one generated by strip clearcutting. Opting for an irregular shelterwood approach would result in two-aged stand structures that contribute to structural rehabilitation over time (Raymond et al., 2009). Complexity enhancement could also include restoring deadwood flow (Keeton, 2006; Ford and Keeton, 2017). Such forest stands would provide diverse habitats for plant and animal species, carbon sequestration and other ecosystem services. Yet we still have little experimental evidence about these alternatives. Future work could include measures to assess how well stands rehabilitated for timber production might fulfill other non-commodity functions and if they require additional treatments to sustain other ecosystem services.

5. Conclusions

Our study assessed an array of silvicultural scenarios that might help to restore functionality for timber production in temperate mixedwood stands. Depending on the rehabilitation goal and initial

conditions, a gradient of passive to active rehabilitation measures might be considered. The recovery of basal area and acceptable growing stock in absence of treatment indicates the resilient nature of these mixedwood stands, especially when conifers are present in regenerating layers. Yet if the goal is to speed the improvement of stand composition, quality and structure, then active rehabilitation may be required. Shelterwood systems combined to scarification (raking) during a good seed crop would improve stand quality, while managing for compositional and structural objectives. In addition, maintaining a component of spruces and other conifers in the overstory and understory would accelerate recovery of basal area and acceptable growing stock, and maintain both species diversity and a seed source for conifer regeneration. Where seed sources and advance conifer regeneration are lacking, enrichment planting may be necessary to maintain the mixedwood composition over time. However, with broader rehabilitation goals, some other components of ecosystem functioning might still be lacking or would recover very slowly (e.g. replenishing of companion species, deadwood and other structural attributes) and require supplemental measures to ensure a desirable outcome.

CRedit authorship contribution statement

Patricia Raymond: Conceptualization, Formal analysis, Writing - original draft, Visualization. **Marcel Prévost:** Conceptualization, Methodology. **Vincent Roy:** Conceptualization, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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